

surface area of solid of revolution

When an arc of a curve revolves about a fixed line, a surface of revolution is formed.

The area of the surface of solid obtained by revolving about the x-axis, the arc of the curve $y = f(x)$ intercepted between the points whose x-axis are a and b is given

by
$$S = \int_a^b 2\pi y \, ds$$

(i) formula in cartesian form

Area of the curved surface of the solid generated by revolving the area bounded by the curve $y = f(x)$, x-axis and the ordinates at $x = a$ and $x = b$ about the x-axis is given by

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

(ii) formula in parametric form

The area of the curved surface of the solid generated by revolution of the area bounded by the curve $x = f(t)$, $y = g(t)$, the x-axis and the ordinates $t = a$ and $t = b$ about x-axis is given by

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \, dt$$

(iii) formula in polar form

The area of the curved surface of the solid of revolution of the area

bounded by the curve $r = f(\theta)$ and the radii vectors $\theta = \alpha$ and $\theta = \beta$ about the initial line is given by

$$S = \int_{\alpha}^{\beta} r \sin \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Example

(1) Find the area of the surface generated by revolving about the axis of x : the arc of the parabola $y^2 = 4ax$ from origin to the point where $x = a$

Equation of the given parabola is $y^2 = 4ax$

Differentiating w.r. to x

$$2y \frac{dy}{dx} = 4a, \quad \frac{dy}{dx} = \frac{4a}{2y} = \frac{2a}{y}$$

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{4a^2}{y^2} = 1 + \frac{4a^2}{4ax}$$

$$= 1 + \frac{a}{x} = \frac{x+a}{x}$$

Required surface area

$$S = \int_0^a 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= 2\pi \int_0^a 2\sqrt{ax} \sqrt{\frac{x+a}{x}} dx$$

$$= 4\pi\sqrt{a} \int_0^a \sqrt{x+a} dx$$

$$= 4\pi\sqrt{a} \left[\frac{(x+a)^{3/2}}{3/2} \right]_0^a$$

$$= \frac{8}{3}\pi\sqrt{a} \left[(2a)^{3/2} - a^{3/2} \right]$$

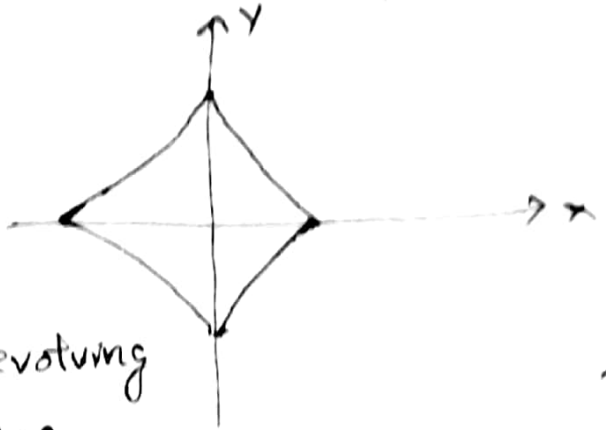
$$= \frac{8}{3}\pi\sqrt{a} \cdot a^{3/2} (2\sqrt{2} - 1)$$

$$= \frac{8(2\sqrt{2}-1)}{3} \pi a^2$$

(1) Show that the surface area of the solid generated by revolving the curve
 $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ (i.e. $x^{2/3} + y^{2/3} = a^{2/3}$)

This curve is symmetrical about x-axis and y-axis

Hence the required area is double the surface of the solid generated by revolving portion of the curve in the first quadrant



For the portion of the curve in the first quadrant θ varies from 0 to $\frac{\pi}{2}$

$$\frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta, \quad \frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

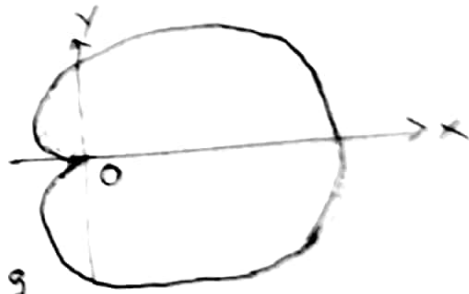
$$\begin{aligned} \therefore \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} &= \sqrt{9a^2 \cos^4 \theta \sin^2 \theta + 9a^2 \cos^2 \theta \sin^4 \theta} \\ &= 3a \sqrt{\cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta)} \\ &= 3a \sin \theta \cos \theta \end{aligned}$$

Hence the required surface is given by

$$\begin{aligned} S &= 2 \int_0^{\pi/2} 2\pi y \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\ &= 4\pi \int_0^{\pi/2} (a \sin^3 \theta) (3a \sin \theta \cos \theta) d\theta \\ &= 4\pi \times 3a^2 \int_0^{\pi/2} \sin^4 \theta \cos \theta d\theta \\ &= 12\pi a^2 \left(\frac{\sin^5 \theta}{5} \right)_0^{\pi/2} \\ &= \frac{12}{5} \pi a^2 (\sin^5 \frac{\pi}{2} - \sin^5 0) = \underline{\underline{\frac{12}{5} \pi a^2}} \end{aligned}$$

3) Find the area of the surface of the solid formed by the revolution of the cardioid $r = a(1 + \cos \theta)$ about its line of symmetry

Given curve is symmetric about the initial line OX



Surface area of the solid generated by revolving the whole cardioid is same as the surface area of the solid generated by revolving the upper half of the cardioid. For this upper half of the curve θ varies from 0 to π

We have $r = a(1 + \cos \theta)$

$$\frac{dr}{d\theta} = -a \sin \theta$$

$$\begin{aligned} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} &= \sqrt{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta} \\ &= a \sqrt{1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta} \\ &= a \sqrt{2 + 2 \cos \theta} \\ &= a \sqrt{2(1 + \cos \theta)} = a \sqrt{2 \times 2 \cos^2 \frac{\theta}{2}} \\ &= 2a \cos \frac{\theta}{2} \end{aligned}$$

$\therefore$ Required surface area is given by

$$\begin{aligned} S &= \int_0^\pi 2\pi r \sin \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\ &= 2\pi \int_0^\pi r \sin \theta \cdot 2a \cos \frac{\theta}{2} \\ &= 2\pi \int_0^\pi a(1 + \cos \theta) \sin \theta (2a \cos \frac{\theta}{2}) d\theta \\ &= 4\pi a^2 \int_0^\pi 2 \cos^2 \frac{\theta}{2} \cdot 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \frac{\theta}{2} d\theta \end{aligned}$$

$$= 16 \pi a^2 \int_0^{\pi} \cos^4 \frac{\theta}{2} \sin \frac{\theta}{2} d\theta$$

$$= 32 \pi a^2 \int_0^{\pi/2} \cos^4 \phi \sin \phi d\phi$$

$$= 32 \pi a^2 \left[-\frac{\cos^5 \phi}{5} \right]_0^{\pi/2}$$

$$= 32 \pi a^2 \left(-\frac{1}{5} (0 - 1) \right)$$

$$= \frac{32}{5} \pi a^2$$

Put $\frac{\theta}{2} = \phi$

$d\theta = 2 d\phi$

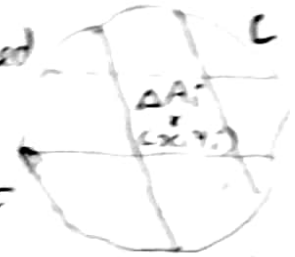
when $\theta = \pi$, $\phi = \frac{\pi}{2}$

when $\theta = 0$, $\phi = 0$

Double Integrals

Let $f(x, y)$ be continuous and single valued function of x and y within region R bounded by a closed curve C

Let the region R be subdivided into n regions of areas $\Delta A_1, \Delta A_2, \dots, \Delta A_n$. Let (x_i, y_i) be any point in the subregion area ΔA_i then double integral of $f(x, y)$ over region R is given by

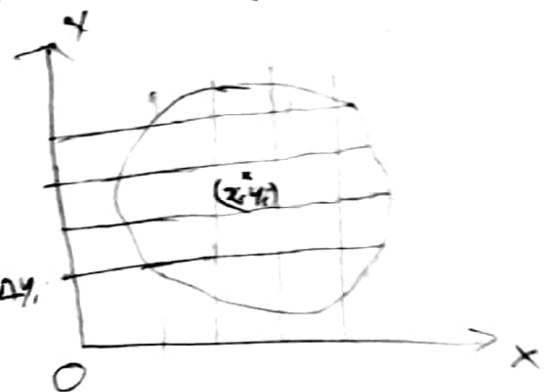


$$\iint_R f(x, y) dA = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i, y_i) \Delta A_i$$

In order to simplify evaluation of the double integral we often consider subregions R_i as rectangular grid obtained by drawing lines parallel to co-ordinate axes

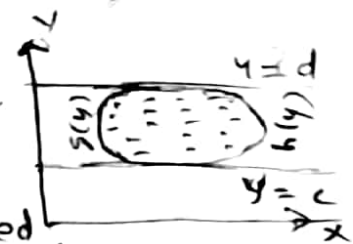
$$\Delta A_i = \text{Area of rectangle } R_i = \Delta x_i \cdot \Delta y_i$$

$$\therefore \iint_R f(x, y) dx dy = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i, y_i) \Delta x_i \Delta y_i$$



Note 1 Suppose region R is described by the inequalities $c \leq y \leq d$ and $g(y) \leq x \leq h(y)$

$$\iint_R f(x, y) dx dy = \int_{y=c}^{y=d} \int_{x=g(y)}^{x=h(y)} f(x, y) dx dy$$



Note 2 If the region R is described by $a \leq x \leq b$ and $r(x) \leq y \leq s(x)$

$$\iint_R f(x, y) dx dy = \int_{x=a}^{x=b} \int_{y=r(x)}^{y=s(x)} f(x, y) dy dx$$



(1) Evaluate $\int_0^2 \int_1^2 (x^2 + y^2) dx dy$

$$I = \int_0^2 \int_1^2 (x^2 + y^2) dx dy$$

$$= \int_0^2 \left[\int_1^2 (x^2 + y^2) dx \right] dy$$

$$= \int_0^2 \left(\frac{x^3}{3} + y^2 x \right)_1^2 dy$$

$$= \int_0^2 \left(\frac{8}{3} + 2y^2 \right) - \left(\frac{1}{3} + y^2 \right) dy$$

$$= \int_0^2 \left(\frac{7}{3} + y^2 \right) dy = \left(\frac{7}{3} y + \frac{y^3}{3} \right)_0^2$$

$$= \frac{14}{3} + \frac{8}{3} = \frac{22}{3}$$

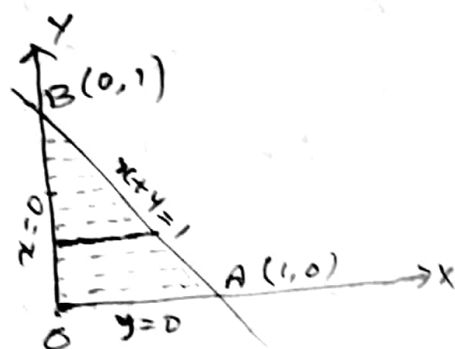
First integrate $x^2 + y^2$ w.r. to x between the limits $x=1$ and $x=2$

Integrate w.r. to y between the limits $y=0$ and $y=2$

(2) Evaluate $\iint_S (x^2 + y^2) dx dy$ where S is the region

in which each of $x, y \geq 0$ and $x + y \leq 1$

Region of integration is the triangle formed by the lines $x=0$ (i.e. y axis), $y=0$ (i.e. x axis) and $x+y=1$



In this region y varies from 0 to 1
for each value of y , x varies from 0 to $1-y$.

$$\iint_S (x^2 + y^2) dx dy = \int_0^1 \int_0^{1-y} (x^2 + y^2) dx dy$$

$$= \int_0^1 \left[\int_0^{1-y} (x^2 + y^2) dx \right] dy$$

$$= \int_0^1 \left(\frac{x^3}{3} + y^2 x \right)_0^{1-y} dy$$

$$\begin{aligned}
 &= \int_0^1 \left[\frac{(1-y)^3}{3} + (1-y)y^2 \right] dy \\
 &= \frac{1}{3} \int_0^1 \left[(1-3y+3y^2-y^3) + 3(y^2-y^3) \right] dy \\
 &= \frac{1}{3} \int_0^1 (1-3y+3y^2-y^3+3y^2-3y^3) dy \\
 &= \frac{1}{3} \int_0^1 (1-3y+6y^2-4y^3) dy \\
 &= \frac{1}{3} \left[y - \frac{3y^2}{2} + 6\frac{y^3}{3} - 4\frac{y^4}{4} \right]_0^1 \\
 &= \frac{1}{3} \left(1 - \frac{3}{2} + 2 - 1 \right) = \frac{1}{3} \times \frac{1}{2} = \underline{\underline{\frac{1}{6}}}
 \end{aligned}$$

- (3) Evaluate the double integral $\iint xy(x+y) dx dy$ over the region bounded by the $y=x^2$ and the line $y=x$

The parabola $y=x^2$ and the line $y=x$ meet at $(0,0)$ and $(1,1)$. Region of integration



is as shown in the figure

$$I = \int_0^1 \int_y^{\sqrt{y}} (xy^2 + x^2y) dx dy$$

In this region y varies from 0 to 1. For each value of y , x varies from y to $\sqrt{y}$

$$\begin{aligned}
 &= \int_0^1 \left(\frac{x^2}{2} y^2 + \frac{x^3}{3} y \right) \Big|_y^{\sqrt{y}} dy \\
 &= \int_0^1 \left(\left(\frac{y^3}{2} + \frac{(\sqrt{y})^3 y}{3} \right) - \left(\frac{y^4}{3} + \frac{y^4}{2} \right) \right) dy = \int_0^1 \left(\frac{y^3}{2} + \frac{y^{5/2}}{3} - \frac{y^4}{3} + \frac{y^4}{2} \right) dy \\
 &= \left(\frac{y^4}{8} + \frac{y^{7/2}}{7/2 \times 3} - \left(\frac{y^5}{15} + \frac{y^5}{10} \right) \right) \Big|_0^1 = \frac{1}{8} + \frac{2}{21} - \frac{1}{15} - \frac{1}{10} = \underline{\underline{\frac{3}{56}}}
 \end{aligned}$$

changing order of integration

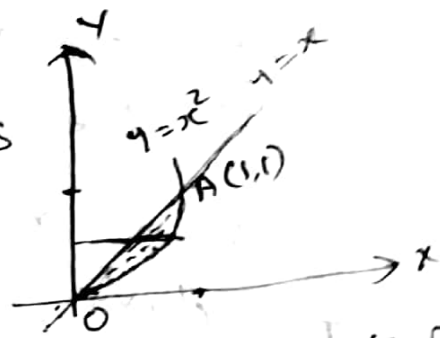
In double integral if both limits are constants for each of the variables, the order of integration is immaterial. If limits of integration are variables, to change the order of integration first sketch the region of integration and from it determine the new limits of the variables.

Problems

- (1) change the order of integration in $\int_0^1 \int_{x^2}^x (2x^2 + y) dy dx$ and hence evaluate it.

In this integral first integration is w.r. to y between limits $y = x^2$ and $y = x$. Second integration is w.r. to x between the limits $x = 0$ and $x = 1$.

Region of integration is as shown in the figure.
changing the order



y varies from 0 to 1
For any value of y , x varies from y to $\sqrt{y}$

$$\begin{aligned} \therefore I &= \int_0^1 \int_{x^2}^x (2x^2 + y) dy dx \\ &= \int_0^1 \int_y^{\sqrt{y}} (2x^2 + y) dx dy \\ &= \int_0^1 \left(\frac{2x^3}{3} + yx \right)_y^{\sqrt{y}} dy \end{aligned}$$

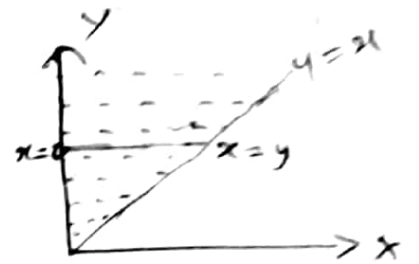
$$\begin{aligned}
&= \int_0^1 \left[\frac{2}{3} (y^{3/2} - y^3) + y(\sqrt{y} - y) \right] dy \\
&= \int_0^1 \left(\frac{2}{3} y^{3/2} - \frac{2}{3} y^3 + y^{3/2} - y^2 \right) dy \\
&= \int_0^1 \left(\frac{5}{3} y^{3/2} - \frac{2}{3} y^3 - y^2 \right) dy \\
&= \left(\frac{5}{3} \frac{y^{5/2}}{5/2} - \frac{2}{3} \frac{y^4}{4} - \frac{y^3}{3} \right) \Big|_0^1 \\
&= \frac{2}{3} - \frac{2}{12} - \frac{1}{3} = \frac{8-2-4}{12} = \underline{\underline{\frac{1}{6}}}
\end{aligned}$$

(2) change the order of integration in

$$\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx \text{ and hence evaluate it.}$$

In this integral y varies from x to ∞ and x from 0 to ∞

changing the order of integration



y varies from 0 to ∞

and ~~x from~~ for any value of y , x varies from 0 to y

$$I = \int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx = \int_0^\infty \int_0^y \frac{e^{-y}}{y} dx dy$$

$$= \int_0^\infty \frac{e^{-y}}{y} (x)_0^y dy = \int_0^\infty \frac{e^{-y}}{y} \cdot y dy$$

$$= \int_0^\infty e^{-y} dy = \left(\frac{e^{-y}}{-1} \right)_0^\infty = \frac{1}{-1} (0-1) \quad \left(\begin{array}{l} e^{-\infty} = 0 \\ e^0 = 1 \end{array} \right)$$

$$= \underline{\underline{1}}$$

Area of a region in a plane

$$\text{Area of a region in 2D space} \\ = \iint_R dx dy$$

Example Find by double integration area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Because of symmetry,
area of the ellipse is 4 times
its area in first quadrant



In the first quad y varies from 0 to b

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{a^2} = 1 - \frac{y^2}{b^2} = \frac{b^2 - y^2}{b^2}$$

$$\Rightarrow x^2 = \frac{a^2}{b^2} (b^2 - y^2)$$

$$\Rightarrow x = \frac{a}{b} \sqrt{b^2 - y^2}$$

$\therefore x$ varies from 0 to $\frac{a}{b} \sqrt{b^2 - y^2}$

$$\therefore \text{Area of the ellipse} = 4 \int_0^b \int_0^{\frac{a}{b} \sqrt{b^2 - y^2}} dx dy$$

$$= 4 \int_0^b (x) \Big|_0^{\frac{a}{b} \sqrt{b^2 - y^2}} dy$$

$$= 4 \int_0^b \frac{a}{b} \sqrt{b^2 - y^2} dy$$

$$= \frac{4a}{b} \int_0^b \sqrt{b^2 - y^2} dy$$

$$= \frac{4a}{b} \left[\frac{y}{2} \sqrt{b^2 - y^2} + \frac{b^2}{2} \sin^{-1} \left(\frac{y}{b} \right) \right]_0^b$$

$$= \frac{4a}{b} \left[0 + \frac{b^2}{2} \sin^{-1}(1) - (0 + 0) \right]$$

$$= \frac{4a}{b} \cdot \frac{b^2}{2} \cdot \frac{\pi}{2} = \underline{\underline{\pi ab}}$$

Problem

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Evaluate $\iint_R r^3 dr d\theta$ over the area bounded ~~by~~ between the circles $r = 2 \cos \theta$ and $r = 4 \cos \theta$

$$r = 2 \cos \theta \Rightarrow r^2 = 2r \cos \theta$$

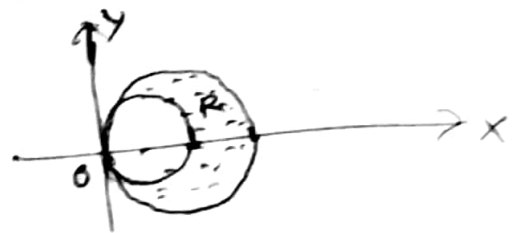
$$\Rightarrow x^2 + y^2 = 2x$$

$$\Rightarrow x^2 - 2x + y^2 = 0$$

$$\Rightarrow (x-1)^2 + y^2 = 1$$

It is a circle with centre $(1, 0)$ and radius 1 unit.
ii) $r = 4 \cos \theta \Rightarrow (x-2)^2 + y^2 = 4$ which is a circle with centre $(2, 0)$ and radius 2 units.

shaded portion is the required region. In this region r varies from $2 \cos \theta$



to $4 \cos \theta$ and θ varies from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$

$$I = \iint_R r^3 dr d\theta = \int_{-\pi/2}^{\pi/2} \int_{2 \cos \theta}^{4 \cos \theta} r^3 dr d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \left(\frac{r^4}{4} \right)_{2 \cos \theta}^{4 \cos \theta} d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \frac{1}{4} (256 \cos^4 \theta - 16 \cos^4 \theta) d\theta$$

$$= \frac{1}{4} \times 2 \int_0^{\pi/2} (240 \cos^4 \theta) d\theta$$

$$= \frac{1}{2} \times 240 \left(\int_0^{\pi/2} \cos^4 \theta d\theta \right)$$

$$= 120 \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \underline{\underline{\frac{45\pi}{2}}}$$